

PAIMANA SENTINEL

Problem Statement ID : 26103

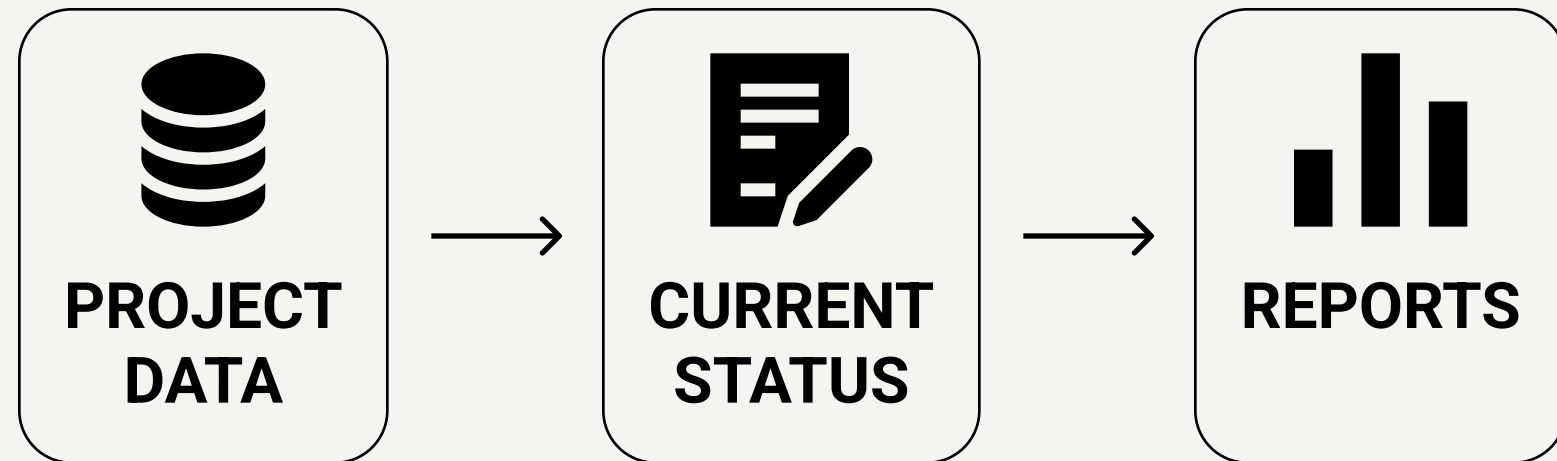
Problem Statement Title : Use case on web-based integrated project-monitoring platform

Theme : Smart Automation

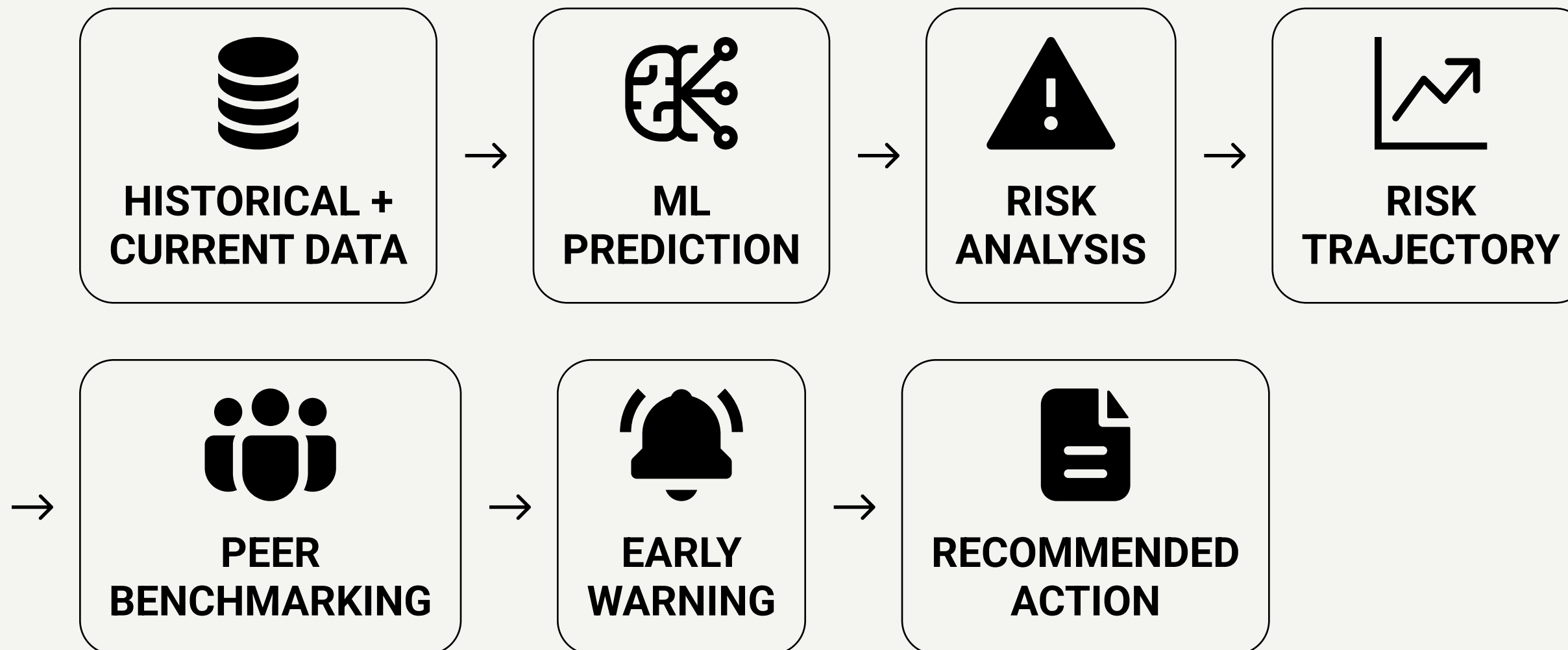
PS Category : Software

Team Name : Deadlock

EXISTING MONITORING

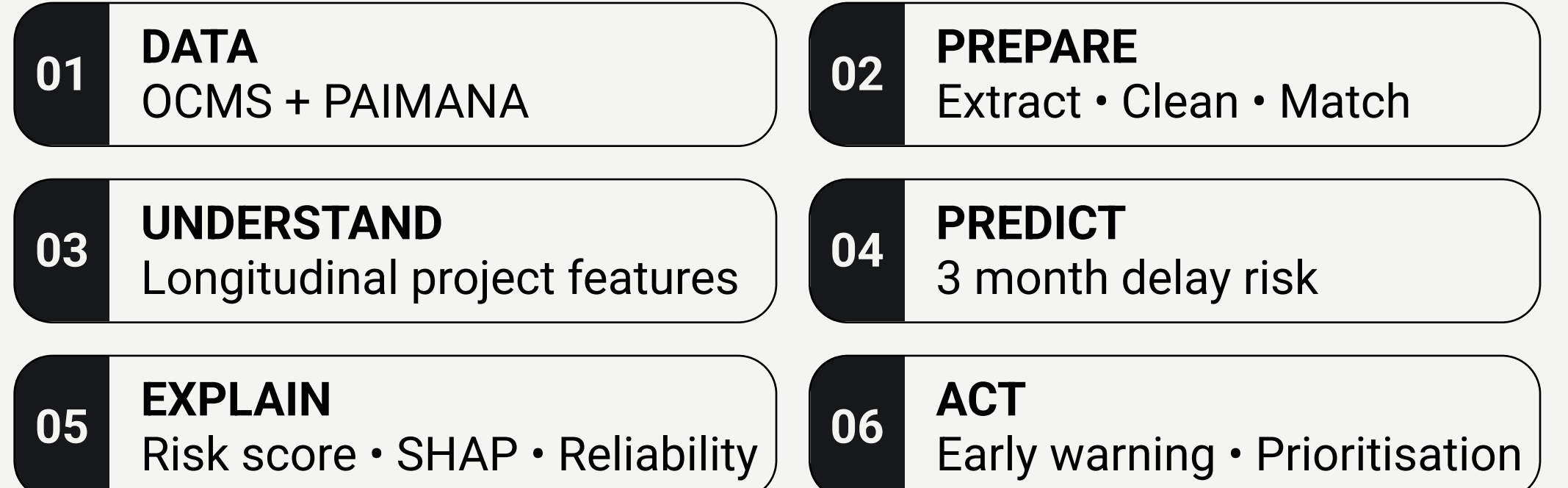


PREDICTIVE MONITORING



PROPOSED SOLUTION

LLM integrated predictive monitoring for PAIMANA



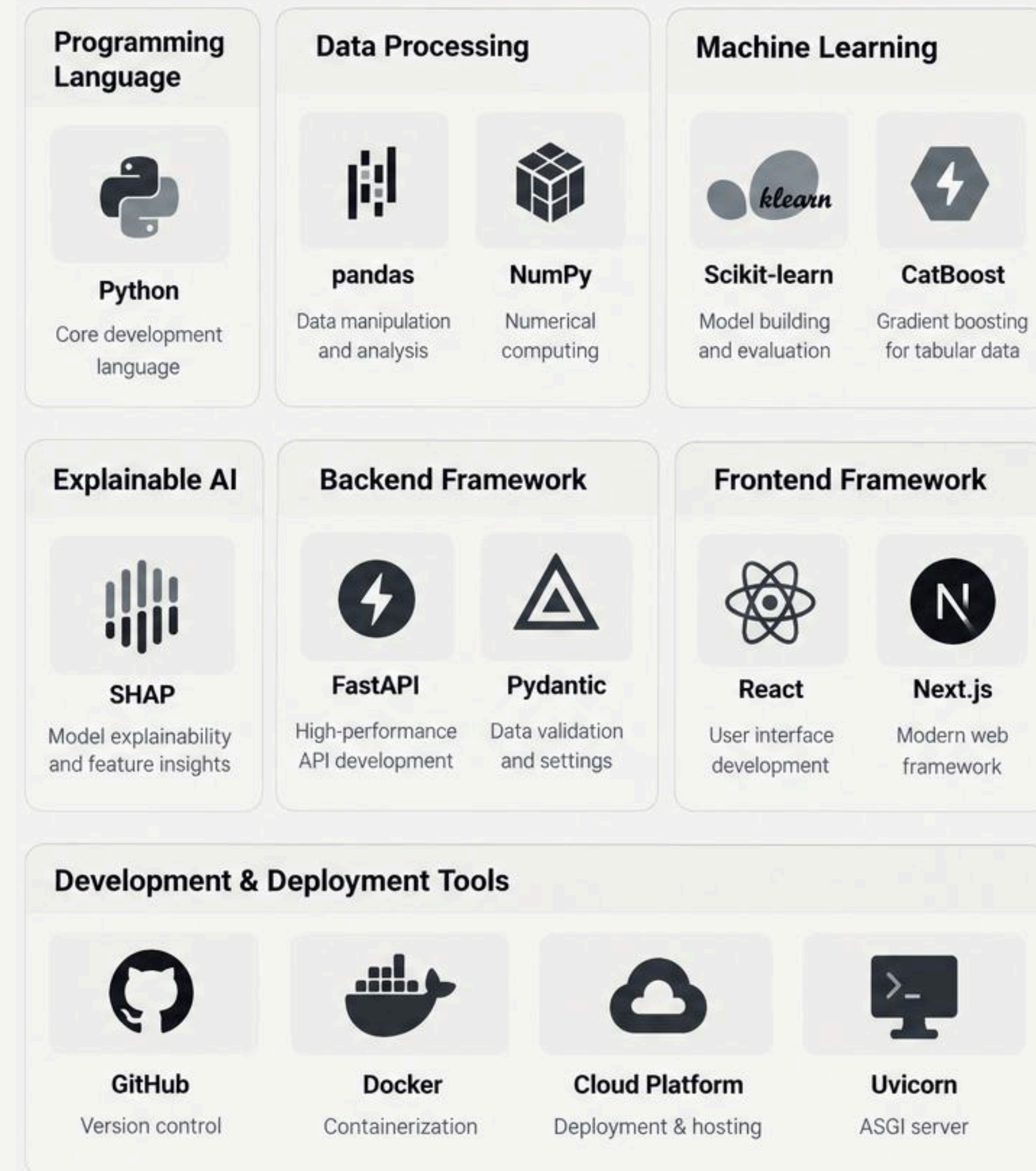
INNOVATION AND UNIQUENESS

- 3 month schedule delay prediction
- Dynamic risk trajectory
- Peer benchmarking
- Explainable AI with SHAP
- Data reliability & confidence scoring
- Early warning & intervention prioritisation

METHODOLOGY / IMPLEMENTATION



TECHNOLOGY STACK



FEASIBILITY

An intelligence layer, not a replacement

PAIMANA

Regularly updated project information and existing monitoring infrastructure



PAIMANA SENTINEL

Predictive intelligence layered on top of the existing system

CHALLENGES

Where predictions can fail

Model reliability depends on the quality, consistency and completeness of project data and its ability to capture real world factors.

DATA QUALITY

Incomplete or inconsistent records

SECTOR VARIATION

Different patterns across sectors

MODEL UNCERTAINTY

False warnings or unreliable predictions

MITIGATION

Trust is built into the system

Multiple safeguards make predictions more interpretable and help officials understand when to trust the model.

DATA CHECKS + PROJECT MATCHING

Reliability indicators + SHAP explanations + human led decisions

COST OVERRUN



Original Approved Cost (in cr.)

₹ 37,12,662



Latest Revised Cost (in cr.)

₹ 42,78,402

POTENTIAL IMPACT ON THE TARGET AUDIENCE

Enables proactive, data driven infrastructure monitoring for IPMD, MoSPI / PAIMANA, helping identify high risk projects early and prioritise timely interventions.

BENEFITS OF THE SOLUTION

SOCIAL

- Faster identification of delayed projects
- Timely delivery of public infrastructure
- Improved transparency and accountability

ECONOMIC

- Reduces project delays
- Optimises public resources
- Improves execution efficiency

GOVERNANCE

- Shifts descriptive to predictive monitoring
- Enables early warnings and action
- Supports peer benchmarking & explainable decision making

TECHNICAL

- Converts PAIMANA / OCMS data into actionable insights
- Provides risk scoring and risk trajectories
- Provides project level explanations and data reliability
- Enables peer based project comparison

Research And References

PRIMARY SOURCE

- 01 PAIMANA PROJECT MONITORING**
<https://paimana-proj.mospi.gov.in/ProjectMonitoring>
- 02 PAIMANA REPORTS**
<https://paimana-proj.mospi.gov.in/ReportPage>
- 03 PAIMANA PUBLIC DASHBOARD**
<https://paimana-proj.mospi.gov.in/Home/PublicDashboardNew>

TECHNICAL RESEARCH AREAS

- Predictive analytics
- Longitudinal project monitoring
- Schedule delay prediction
- Machine learning for tabular data
- Explainable AI / SHAP
- Peer benchmarking
- Early warning systems
- Decision support systems

PROJECT RESEARCH

- OCMS and PAIMANA data integration
- Project identity matching
- Data quality and anomaly detection
- Longitudinal feature engineering
- Leakage safe model evaluation
- Risk prediction and calibration
- SHAP based explanations
- Risk trajectory analysis
- Peer benchmarking